



How data, AI, and ML are reshaping the financial services industry

AWS Partners drive generative AI innovation to transform the financial sector

Table of contents

Rethink financial intelligence	3
Accelerate innovation with AWS Partners.....	4
Deloitte: Moving generative AI from POC to production.....	5
NVIDIA: Revolutionizing financial services with AI compute power and AWS	6
Splunk: Mitigating financial risks with observability and security	7
Databricks: Enhancing financial services through data intelligence	8
EPAM: Improving financial decision-making with rapid data access	8
Genesys Cloud Services Inc.: Innovating security solutions with AI.....	9
HCLTech: Driving tech adoption for finance.....	9
Informatica: Enhancing data management with AI.....	10
MongoDB: Boosting fraud detection with data	10
Smash: Integrating AI for stronger financial compliance	11
Immuta: Securing and optimizing AI analytics	11
Infosys: Accelerating solutions with AI tools	12
Neo4j: Enhancing AI accuracy with graph technology.....	12
Vertical Relevance: Fast-tracking deployments with AI.....	13
Virtusa: Pioneering AI-driven transformation for intelligent enterprise	13
Transform your financial services strategy	14

Rethink financial intelligence

Highlights from the AWS 2024 Financial Services Symposium

Advancements in artificial intelligence (AI), machine learning (ML), and generative AI are enabling financial services organizations to extract game-changing insights from data, automate complex processes, and create hyper-personalized customer experiences. But while data has never wielded such power, these advancements are not without challenges. The Amazon Web Services (AWS) Financial Services Symposium brought together senior executives across banking, payments, capital markets, and insurance to explore how they are working with AWS to use data responsibly to better engage customers, make smarter business decisions, and fortify critical applications.

At the Financial Services Symposium, we explored one key question: How?

- How can financial services organizations choose the right AWS service and AWS Partner to address their specific business challenges, requirements, or opportunities?
- How can they make those services work with their existing technology investments?
- How can they navigate the choices that come with executing transformational change and catalyzing innovation?

The solutions to these challenges often lie in the powerful combination of human and machine intelligence.

With AI, machines can now complete complex, time-consuming tasks in a fraction of the time that humans could. But humans still have the upper hand in thinking strategically and identifying problems that technology can help solve. AWS is helping the financial services industry use AI to solve these problems by providing capabilities that combine human and machine intelligence to drive meaningful transformation.

It's time to rethink intelligence. Human imagination has never been more important in identifying the right problems to solve to improve the financial system for consumers and businesses alike. In this eBook, learn how financial services institutions are working with AWS and AWS Partners to leverage human intelligence, AI, and ML to spur innovation, from crafting tailored customer interactions to accelerating decision-making to protecting their most critical information.

Watch theCUBE interview keynote recap with Scott Mullins, AWS Financial Services General Manager ›
Watch the New York AWS Financial Services Symposium on-demand ›



Accelerate innovation with AWS Partners

AWS offers a robust suite of services tailored to meet the specific needs of the financial services industry. These services, combined with AWS support for compliance certifications such as PCI DSS, SOC, and FedRAMP, help financial institutions achieve scalability and security goals while accelerating innovation.

AWS Marketplace provides easy access to transformative, cloud-based, and purpose-built solutions for the financial services industry. By simplifying vendor management through a curated catalog of third-party offerings with global reach, AWS Marketplace helps financial services institutions improve service delivery across regions, speed time to market, and unlock the potential of their data.

AWS Financial Services and Generative AI Competency Partners leverage their deep industry knowledge and cutting-edge cloud technology to propel financial institutions forward. These experts are indispensable allies when it comes to navigating the complexities of the financial sector. For example, AWS Partners enable financial services institutions to leverage the power of generative AI and other new technologies at scale, driving unmatched agility and innovation.

Read on to discover how AWS Partners empower organizations to use data to drive innovation and transformation in today's dynamic financial landscape.





Moving generative AI from POC to production

Financial services firms are building generative AI demos and proofs of concept (POCs) as they explore the most effective use cases for their business needs. Moving these POCs to scalable, production-grade applications that are widely used and deliver value requires a level of rigor and AI engineering skills typically not seen in enterprises today.

There is a critical distinction between crafting impressive demos and building robust, production-grade applications. While demos can be created in as little as one to three weeks, the transition to production necessitates a strategic shift in focus. It becomes critical to master the following four AI engineering disciplines for production deployments of generative AI applications:

1. Evaluation: Process and test to evaluate the fit of a large language model (LLM) for a problem.
2. Guardrails: Ensure that the LLM answers in a manner aligned with the organization's policies.
3. Defensive UI: UI to be the feedback and data labeling mechanism.
4. LLMOps: Deploy and monitor LLMs in production in a consistent manner.

For example, through structured AI policy development and innovative use-case workshops, Deloitte enabled a large financial firm to accelerate POCs to applications in production. Deloitte focused on guiding its client through AI engineering challenges, such as developing and fine-tuning LLMs for practical use cases. The team leveraged iterative prompt engineering and tools like [Amazon Bedrock](#) and [Amazon SageMaker Ground Truth](#) to increase response accuracy from 70 to 97 percent.

With Deloitte's partnership, the financial firm effectively transitioned generative AI applications from POC to production. The firm started with 20 POCs, successfully moving three into production within 12 months, with plans for over 10 more by year-end. It also saw dramatic improvements in efficiency as the result of implementing AI and automation solutions that significantly streamlined processes and reduced manual effort.

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Lack of adequate governance is one of the biggest blockers for AI deployments. Using Amazon AI guardrails helped enforce responsible AI.

Niraj Juneja

AI and ML Specialist Leader for
Financial Services, Deloitte

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NVIDIA's accelerated computing platform, along with AWS, optimizes for GPU acceleration to reduce costs and improve speed and efficiency of operations for financial institutions.

Malcolm DeMayo

Vice President of Financial
Services, NVIDIA

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Revolutionizing financial services with AI compute power and AWS

With computing demands increasing quickly, the financial services industry needs the power of generative AI and advanced computing. Traditional CPU-based computing platforms are often insufficient for modern AI-driven workloads, and organizations struggle to efficiently process vast amounts of data and train AI models. NVIDIA, a leader in AI computing, offers an Accelerated Compute platform that uses a combination of GPUs, CPUs, DPUs, and specialized software to enhance computational power.

A significant component of NVIDIA's strategy is its AI Factory—a reference architecture designed to centralize and streamline AI development within organizations. This AI Factory, powered by AWS, comprises three main elements: NVIDIA's DGX Cloud, open-source models, and NVIDIA's proprietary models, which are built using the NeMo framework. The AI Factory aims to standardize AI efforts within firms, allowing for efficient experimentation with and deployment of AI models.

NVIDIA's solutions have proven to be transformative for financial services firms. For example, inefficiencies in data-processing workflows increases costs and limits the effectiveness of AI models. NVIDIA's accelerated computing platform, along with AWS, optimizes for GPU acceleration to reduce costs and improve speed and efficiency of operations for financial institutions.

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Mitigating financial risks with observability and security

Financial services companies must prioritize observability, security, and compliance to avoid potential risks like data breaches, security vulnerabilities, and incidents compromising personal information. In addition, new government mandates and regulatory requirements are continuously emerging, necessitating a robust compliance strategy to mitigate disruption. Complicating matters further, companies must address these evolving threats while maintaining digital resilience and operational agility.

Splunk, a big data platform, helps customers integrate observability and security practices to enhance digital resilience. This involves monitoring IT infrastructures and applications to gain real-time insights into system performance and potential issues, with added security that protects data and prevents breaches. Splunk processes vast amounts of data—from sources such as metrics, logs, and traces—to provide comprehensive analytics. Splunk's approach includes using OpenTelemetry to standardize data collection, improving speed and scalability. This helps organizations transition from basic investigative searches to proactive monitoring and situational awareness. Splunk also leverages AI capabilities to help with anomaly detection, predictive analysis, and automation of routine and repetitive tasks to promote efficiency.

Nasdaq, the US-based stock market exchange, started using Splunk in 2010 for basic security-event monitoring. Over time, it expanded usage to include proactive monitoring, situational awareness, and business analytics, processing terabytes of data quickly. This evolution allowed Nasdaq to improve its operational agility and security posture significantly. With Splunk, Nasdaq was able to reduce resolution time from days to minutes, so that even during high-pressure situations like production outages, problems are swiftly addressed.

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Solving a problem used to take days. With better observability tools, we reduced resolution times to just minutes.

Nimish Doshi

Director of Technical Advisor
Services, Splunk

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Enhancing financial services through data intelligence

The financial services industry faces challenges with managing legacy technology, fragmented customer experiences, regulatory constraints, and high personnel costs. As banks aim to become more instant, invisible, and available, AI plays a crucial role in enhancing experiences and improving access to data.

Databricks, a data intelligence platform, transforms technology infrastructures into cloud-native, data-centric environments. It uses generative AI to personalize customer experiences, democratizes data access for employees, and integrates data processing and analytics into a unified platform. By using AI to simplify data access and analytics, Databricks makes it easier for non-technical users to generate insights from data. This approach empowers users across an organization to derive relevant insights efficiently and boost productivity through AI-enhanced automation.

Junta Nakai, global vice president of financial services and cybersecurity for Databricks, said, “The two most important assets that any financial services company has today are its data and its people.” By enabling a broader range of employees to leverage data in their decision-making processes, Databricks’ approach fosters a truly data-driven culture.

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Improving financial decision-making with rapid data access

Financial services institutions must handle and process vast amounts of data to make informed decisions. Traditional infrastructures often fail to provide data insights promptly due to the sheer volume and complexity of the information.

EPAM, a software engineering and services company, developed Energy Studio: Impact, a cloud-based application that streamlines data processing for S&P Global Commodity Insights. The solution helps S&P Global Commodity Insights, a financial information and analytics company, perform rapid decision-making with large amounts of data. EPAM utilized AWS services to build a scalable, serverless architecture, with Amazon SageMaker playing a crucial role in ML model deployment and real-time inference. The project addressed the challenge of handling large geospatial data sets, standardizing them into a common reference system and delivering high-performance analytics using a GPU-based solution. Energy Studio: Impact enabled users to quickly access, analyze, and visualize data, enhancing client retention, increasing revenue, expanding market reach, and facilitating collaboration.

“EPAM’s solution was transformative for us. It accelerated our ability to turn vast amounts of data into actionable insights, empowering our clients to make informed decisions quickly,” said Andrea Anderson, executive director of digital products at S&P Global Commodities Insights.

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Innovating security solutions with AI

Banks and insurers have a lot of data to protect and must maintain security and compliance while innovating. Migrating from on-premises to cloud technologies can be complicated due to infrastructure size and scale, regulations, and required approval processes.

Genesys Cloud Services Inc. provides client-connectivity technologies that integrate multiple communication platforms. Services such as agent assistance and predictive routing and engagement improve customer experiences and streamline operations using AI and cloud capabilities running on AWS. Integrating these technologies lets banks offer more sophisticated and secure services, including in-app secure calls, video interactions, and real-time assistance.

An early adopter of [Amazon Bedrock](#), Genesys helped a European bank develop an in-app contact solution featuring predictive engagement software that uses AI to anticipate customer needs. The software predicts why a customer is calling and proactively presents relevant information, reducing the need for live agent intervention and speeding up resolution times. Supporting over 15,000 users across more than 80 departments, the bank's contact center found resiliency and agility running on the Genesys Cloud platform powered by AWS. The AI-driven solution from Genesys improved the bank's customer satisfaction by 90 percent, increased productivity by 25 percent, and saved \$2.2 million in hardware and IT costs.

HCLTech

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Driving tech adoption for finance

Innovating within the highly regulated financial services industry is inherently challenging due to strict regulations, data privacy requirements and intellectual property concerns. This often results in prolonged timelines for new technologies to transition from proof-of-concept (POC) stages to large-scale production.

HCLTech, a premier consulting and AWS Financial Services Competency partner, addresses these challenges by leveraging their combined strengths to drive innovation while ensuring compliance. HCLTech also supports continuous re-skilling and organizational change management, with training programs showing successful implementations and providing hands-on learning experiences. "We've built labs where customers can understand what the options are from a generative AI, cloud and data point of view to show how use cases can be brought to life," said Anubhav Mehrotra from HCLTech.

By helping to transition generative AI applications from POC to full-scale production, HCLTech improves efficiency and reduces costs for financial services organizations, such as Amerisure Insurance. This leads to a more seamless and innovative customer experience and frees up staff to focus on strategic initiatives.



Enhancing data management with AI

When it comes to generative AI and digital transformation, there's an increasing demand for data readiness. Even where there's significant investment in AI, the foundational requirement of reliable, accessible, and governed data often lags behind, leading to inefficiencies and missed opportunities. In addition, outdated tools and fragmented processes make it difficult to scale AI effectively.

Informatica helps modernize data management and governance practices by consolidating and standardizing processes, leveraging modern technologies to ensure data is available, trustworthy, and usable across an entire organization. Informatica helps companies leverage AI with AI-driven platform enhancements like CLAIRE, an AI copilot powered by AWS that uses ML and natural language processing to automate data management tasks. This can unlock new levels of operational efficiency, accelerate time to value for AI initiatives, and democratize data management across business functions to reduce costs and risks associated with legacy approaches. Informatica integrates seamlessly with cloud services to enhance data integration and governance capabilities in cloud ecosystems. This helps organizations prioritize robust data governance and management strategies, ensuring that data remains compatible with and optimized for cloud-based applications and services.

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Boosting fraud detection with data

Competition is intense for financial services firms where data is pivotal—historical analytics have been the norm, but real-time insights are now crucial for competitive advantage. MongoDB's developer data platform offers significant architectural advantages by enabling organizations to securely unify application data (both structured and unstructured) with AI-related data (vectors).

At the heart of MongoDB's developer data platform is MongoDB Atlas. With a unified, scalable, and flexible platform that integrates operational, analytical, and AI data services, organizations can streamline the development of AI-enriched applications. MongoDB integrates with AI/ML platforms to efficiently process large volumes of data, gaining the actionable intelligence needed for real-time, predictive fraud prevention. The combination of real-time analytics and vector search enables organizations to build intelligent applications powered by semantic search and generative AI over any type of data.

MongoDB Atlas integrates with [Amazon SageMaker Canvas](#) for AI-powered fraud detection and [Amazon Q Developer](#) for greater developer productivity. By combining MongoDB's developer data platform with AWS services like these, as well as [Amazon Bedrock](#) and [Amazon S3](#), financial institutions can swiftly deploy and scale solutions. This helps financial institutions stay ahead in a dynamic market, responding effectively to emerging fraud schemes and customer needs in real time.

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Integrating AI for stronger financial compliance

With increasing regulatory scrutiny across more communication channels, maintaining compliance and detecting misconduct are key for financial services organizations. Compliance teams play a critical role in configuring AI technologies to align with specific regulatory requirements and internal policies. Smarsh's Enterprise Platform helps organizations integrate advanced AI and ML technologies into their compliance programs to streamline processes, improve supervision and monitoring, and gain valuable insights from digital communications.

Smarsh integrates AI for regulatory-grade analytics, ensuring the resiliency, scalability, and security of its Enterprise Platform. Using innovative services like [Amazon EC2](#), [Amazon S3](#), [AWS Key Management Service \(AWS KMS\)](#), and [Amazon Aurora](#), Smarsh can ingest, search, protect, and export content exponentially faster than legacy archives. This includes using AI to sift through vast amounts of data for potential risks like market abuse or fraud. Supplying context and explanations with responses ensures transparency in AI decision-making processes, crucial for meeting regulatory standards. By automating compliance processes and detecting suspicious activities across various communication channels like WhatsApp and voice calls, Smarsh helps financial institutions safeguard against regulatory violations and protect market integrity. This supports current compliance needs and prepares firms to leverage AI for deeper business insights and customer service enhancements in the future.



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Securing and optimizing AI analytics

Banks face challenges managing vast volumes of information while complying with stringent regulations. The Immuta Data Security Platform integrates directly with data platforms such as [Amazon Redshift](#) and [Amazon S3](#), enforcing dynamic policies at the compute and storage levels. These integrations enable compliant and secure self-service analytics at scale, helping some of the world's largest banks automate 95 percent of data access requests and achieve 100 percent audit compliance. Running Immuta on AWS enhances customer service and future readiness in AI-driven analytics leveraging [Amazon SageMaker](#). Immuta helps financial institutions overcome complexity, de-risk data, and harness the full potential of AI.



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Accelerating solutions with AI tools

Financial institutions harness vast amounts of customer data, ripe for AI-driven decision-making. Infosys, a digital services and consulting provider, leverages AI-powered tools to improve results while ensuring decision transparency and regulatory compliance. Infosys collaborates extensively with AWS to deploy AI solutions like Topaz, which utilizes generative AI on [Amazon Bedrock](#) to streamline client service, optimize business processes, and boost productivity. While Topaz focuses on AI, Infosys Cobalt helps customers take advantage of AWS capabilities to accelerate their cloud journey. Cobalt provides access to AWS-powered, cloud-first services, helping businesses co-create new solutions and speed time to market.



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Enhancing AI accuracy with graph technology

Common hurdles when transitioning to generative AI solutions include accuracy and data-backed explanations—necessities in highly regulated sectors. Using graph technology, Neo4j plays a pivotal role in combating fraud and enhancing customer insights within financial services. Neo4j enables building custom knowledge graphs to bolster AI responses with more accurate answers, ensuring trustworthiness in client interactions. To support its clients in modernization, Neo4j aims to enhance scalability and insights through cloud-managed databases and collaboration with AWS. This enables Neo4j to expand its data-driven applications and accelerate deployment, improve data-ingestion capabilities, and enhance scalability for its clients.



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Fast-tracking deployments with AI

The ability to modernize and adopt new technologies can be hindered by compliance issues and legacy systems. Financial services consulting firm Vertical Relevance helps companies adopt AI and leverage cutting-edge technologies to enable comprehensive transformations. Vertical Relevance helped Guardian migrate, modernize, and implement its AWS platform, migrating 80 percent of its applications to AWS. Vertical Relevance also partnered with Guardian to architect and implement its generative AI platform and AI factory to power generative AI business use cases. Guardian can now envision a future where AI empowers all departments and AI initiatives drive business value with revenue-generating use cases.

virtusa

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Pioneering AI-driven transformation for intelligent enterprises

Financial firms need to address various considerations when scaling AI implementations. Digital engineering services provider Virtusa leverages cloud services and capabilities to develop and support Helio, an AI framework, powered by AWS. Helio streamlines data migration and operations to enhance efficiency, and consolidates AI learnings and best practices in a comprehensive package to help clients adopt AI technologies. By 2023, Virtusa facilitated over 60 proofs of concept and established nine successful AI production use cases running on AWS. These helped optimize operations, advance customer experiences, and automate financial services processes.

Transform your financial services strategy

The financial services industry is increasingly adopting AI to increase productivity, improve customer experiences, deliver better insights, and drive innovation. With nearly all financial services organizations utilizing cloud services, understanding how to best leverage AI in the cloud can revolutionize operations, security, and customer experiences.

AWS and AWS Financial Services and Generative AI Competency Partners help deploy generative AI in the cloud to streamline workflows, evolve data accessibility, and improve decision-making processes. These collaborations accelerate digital transformations across banking, payments, insurance, and capital markets, empowering financial services organizations to innovate rapidly and capitalize on new business opportunities.

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